



Advanced Linear Algebra: Foundations to Frontiers

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5.6.2 Summary ¶

The process known as Gaussian elimination is equivalent to computing the LU factorization of the matrix $A \in \mathbb{C}^{m \times m}$

$$: A = LU,$$

where L is a unit lower triangular matrix and U is an upper triangular matrix.

Definition 5.6.2.1. Given a matrix $A \in \mathbb{C}^{m \times n}$ with $m \geq n$, its LU factorization is given by $A = LU$ where $L \in \mathbb{C}^{m \times n}$ is unit lower trapezoidal and $U \in \mathbb{C}^{n \times n}$ is upper triangular with nonzeros on its diagonal.

Definition 5.6.2.2. Principal leading submatrix. For $k \leq n$, the $k \times k$ principal leading submatrix of a matrix A is defined to be the square matrix

$$A_{TL} \in \mathbb{C}^{k \times k} \text{ such that } A = \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right).$$

Lemma 5.6.2.3. Let $L \in \mathbb{C}^{n \times n}$ be a unit lower triangular matrix and $U \in \mathbb{C}^{n \times n}$ be an upper triangular matrix. Then $A = LU$ is nonsingular if and only if U has no zeroes on its diagonal.

Theorem 5.6.2.4. Existence of the LU factorization. Let $A \in \mathbb{C}^{m \times n}$ and $m \geq n$ have linearly independent columns. Then A has a (unique) LU factorization if and only if all its principal leading submatrices are nonsingular.

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A = LU-right-looking(A)
A → ( ATL | ATR
      ABL | ABR )
ATL is 0 × 0
while n(ATL) < n(A)
    ( ATL | ATR
      ABL | ABR ) → ( A00 | a01 | A02
                      a10T | α11 | a12T
                      A20 | a21 | A22 )
    a21 := a21/α11
    A22 := A22 - a21a12T
    ( ATL | ATR
      ABL | ABR ) ← ( A00 | a01 | A02
                      a10T | α11 | a12T
                      A20 | a21 | A22 )
endwhile
    
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Figure 5.6.2.5. Right-looking LU factorization algorithm.

The right-looking algorithm performs the same computations as the algorithm

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for j := 0, ..., n - 1
    for i := j + 1, ..., n - 1
        λi,j := αi,j/αj,j
        αi,j := 0
    endfor
    for i := j + 1, ..., n - 1
        for k = j + 1, ..., n - 1
            αi,k := αi,k - λi,jαj,k
        endfor
    endfor
endfor
    
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compute multipliers

subtract λ_{i,j} times row j from row k

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A = LU-left-looking(A)
A → ( ATL | ATR
      ABL | ABR )
ATL is 0 × 0
while n(ATL) < n(A)
    ( ATL | ATR
      ABL | ABR ) → ( A00 | a01 | A02
                      a10T | α11 | a12T
                      A20 | a21 | A22 )
    Solve L00u01 = a01 overwriting a01 with u01
    α11 := v11 = α11 - a10Ta01
    a21 := a21 - A20a01
    a21 := l21 = a21/α11
    ( ATL | ATR
      ABL | ABR ) ← ( A00 | a01 | A02
                      a10T | α11 | a12T
                      A20 | a21 | A22 )
endwhile
    
```

Figure 5.6.2.6. Left-looking LU factorization algorithm. L_{00} is the unit lower triangular matrix stored in the strictly lower triangular part of A_{00} (with the diagonal implicitly stored).

Solving $Ax = b$ via LU factorization:

- Compute the LU factorization $A = LU$.

- Solve $Lz = b$.
- Solve $Ux = z$.

Cost of LU factorization: Starting with an $m \times n$ matrix A , LU factorization requires approximately $mn^2 - \frac{1}{3}n^3$ flops. If $m = n$ this becomes

$$\frac{2}{3}n^3 \text{ flops.}$$

Definition 5.6.2.7. A matrix L_k of the form

$$L_k = \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & l_{21} & I \end{array} \right),$$

where I_k is the $k \times k$ identity matrix and I is an identity matrix "of appropriate size" is called a Gauss transform.

$$L_k^{-1} = \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & l_{21} & I \end{array} \right)^{-1} = \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & -l_{21} & I \end{array} \right).$$

Definition 5.6.2.8. Given

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix},$$

where $\{\pi_0, \pi_1, \dots, \pi_{n-1}\}$ is a permutation (rearrangement) of the integers $\{0, 1, \dots, n-1\}$, we define the permutation matrix $P(p)$ by

$$P(p) = \begin{pmatrix} e_{\pi_0}^T \\ \vdots \\ e_{\pi_{n-1}}^T \end{pmatrix}.$$

If P is a permutation matrix then $P^{-1} = P^T$.

Definition 5.6.2.9. Elementary pivot matrix. Given $\pi \in \{0, \dots, n-1\}$ define the elementary pivot matrix

$$\tilde{P}(\pi) = \begin{pmatrix} e_{\pi}^T \\ e_1^T \\ \vdots \\ e_{\pi-1}^T \\ e_0^T \\ e_{\pi+1}^T \\ \vdots \\ e_{n-1}^T \end{pmatrix}$$

or, equivalently,

$$\tilde{P}(\pi) = \begin{cases} \begin{pmatrix} I_n & & \\ & & \end{pmatrix} & \text{if } \pi = 0 \\ \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & I_{\pi-1} & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & I_{n-\pi-1} \end{pmatrix} & \text{otherwise,} \end{cases}$$

where n is the size of the permutation matrix.

$[A, p] = \text{LU piv-right-looking}(A)$
$A \rightarrow \left(\begin{array}{c cc} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right), p \rightarrow \begin{pmatrix} p_T \\ p_B \end{pmatrix}$
A_{TL} is 0×0 , p_T has 0 elements
while $n(A_{TL}) < n(A)$
$\left(\begin{array}{c cc} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c cc} A_{00} & a_{01} & A_{02} \\ a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right), \begin{pmatrix} p_T \\ p_B \end{pmatrix} \rightarrow \begin{pmatrix} p_0 \\ \pi_1 \\ p_2 \end{pmatrix}$
$\pi_1 := \max_i \left(\frac{\alpha_{11}}{a_{21}} \right)$
$\left(\begin{array}{c cc} A_{00} & a_{01} & A_{02} \\ a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right) := \left(\begin{array}{c c} I & 0 \\ 0 & P(\pi_1) \end{array} \right) \left(\begin{array}{c cc} A_{00} & a_{01} & A_{02} \\ a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$
$a_{21} := a_{21}/\alpha_{11}$
$A_{22} := A_{22} - a_{21}a_{12}^T$
$\left(\begin{array}{c cc} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c cc} A_{00} & a_{01} & A_{02} \\ a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right), \begin{pmatrix} p_T \\ p_B \end{pmatrix} \leftarrow \begin{pmatrix} p_0 \\ \pi_1 \\ p_2 \end{pmatrix}$
endwhile

Figure 5.6.2.10. Right-looking LU factorization algorithm with partial pivoting.

Solving $Ax = b$ via LU factorization: with row pivoting:

- Compute the LU factorization with pivoting $PA = LU$.
- Apply the row exchanges to the right-hand side: $y = Pb$.
- Solve $Lz = y$.

- Solve $Ux = z$.

Solve $Lz = y$, overwriting y with z (Variant 1)
$L \rightarrow \left(\begin{array}{c c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array} \right), y \rightarrow \left(\begin{array}{c} y_T \\ y_B \end{array} \right)$
L_{TL} is 0×0 and y_T has 0 elements
while $n(L_{TL}) < n(L)$
$\left(\begin{array}{c c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c c c} L_{00} & l_{01} & L_{02} \\ \hline l_{10}^T & \lambda_{11} & l_{12}^T \\ \hline L_{20} & l_{21} & L_{22} \end{array} \right), \left(\begin{array}{c} y_T \\ y_B \end{array} \right) \rightarrow \left(\begin{array}{c} y_0 \\ \psi_1 \\ y_2 \end{array} \right)$
<u>$y_2 := y_2 - \psi_1 l_{21}$</u>
$\left(\begin{array}{c c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c c c} L_{00} & l_{01} & L_{02} \\ \hline l_{10}^T & \lambda_{11} & l_{12}^T \\ \hline L_{20} & l_{21} & L_{22} \end{array} \right), \left(\begin{array}{c} y_T \\ y_B \end{array} \right) \leftarrow \left(\begin{array}{c} y_0 \\ \psi_1 \\ y_2 \end{array} \right)$
endwhile

Figure 5.6.2.11. Lower triangular solve (with unit lower triangular matrix), Variant 1

Solve $Lz = y$, overwriting y with z (Variant 2)
$L \rightarrow \left(\begin{array}{c c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array} \right), y \rightarrow \left(\begin{array}{c} y_T \\ y_B \end{array} \right)$
L_{TL} is 0×0 and y_T has 0 elements
while $n(L_{TL}) < n(L)$
$\left(\begin{array}{c c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c c c} L_{00} & l_{01} & L_{02} \\ \hline l_{10}^T & \lambda_{11} & l_{12}^T \\ \hline L_{20} & l_{21} & L_{22} \end{array} \right), \left(\begin{array}{c} y_T \\ y_B \end{array} \right) \rightarrow \left(\begin{array}{c} y_0 \\ \psi_1 \\ y_2 \end{array} \right)$
<u>$\psi_1 := \psi_1 - l_{10}^T y_0$</u>
$\left(\begin{array}{c c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c c c} L_{00} & l_{01} & L_{02} \\ \hline l_{10}^T & \lambda_{11} & l_{12}^T \\ \hline L_{20} & l_{21} & L_{22} \end{array} \right), \left(\begin{array}{c} y_T \\ y_B \end{array} \right) \leftarrow \left(\begin{array}{c} y_0 \\ \psi_1 \\ y_2 \end{array} \right)$
endwhile

Figure 5.6.2.12. Lower triangular solve (with unit lower triangular matrix), Variant 2

Solve $Ux = z$, overwriting z with x (Variant 1)
$U \rightarrow \left(\begin{array}{c c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right), z \rightarrow \left(\begin{array}{c} z_T \\ z_B \end{array} \right)$
U_{BR} is 0×0 and z_B has 0 elements
while $n(U_{BR}) < n(U)$
$\left(\begin{array}{c c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c c c} U_{00} & u_{01} & U_{02} \\ \hline u_{10}^T & v_{11} & u_{12}^T \\ \hline U_{20} & u_{21} & U_{22} \end{array} \right), \left(\begin{array}{c} z_T \\ z_B \end{array} \right) \rightarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array} \right)$
<u>$\zeta_1 := \zeta_1 / v_{11}$</u>
<u>$z_0 := z_0 - \zeta_1 u_{01}$</u>
$\left(\begin{array}{c c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c c c} U_{00} & u_{01} & U_{02} \\ \hline u_{10}^T & v_{11} & u_{12}^T \\ \hline U_{20} & u_{21} & U_{22} \end{array} \right), \left(\begin{array}{c} z_T \\ z_B \end{array} \right) \leftarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array} \right)$
endwhile

Figure 5.6.2.13. Upper triangular solve Variant 1

Solve $Ux = z$, overwriting z with x (Variant 2)
$U \rightarrow \left(\begin{array}{c c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right), z \rightarrow \left(\begin{array}{c} z_T \\ z_B \end{array} \right)$
U_{BR} is 0×0 and z_B has 0 elements
while $n(U_{BR}) < n(U)$
$\left(\begin{array}{c c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c c c} U_{00} & u_{01} & U_{02} \\ \hline u_{10}^T & v_{11} & u_{12}^T \\ \hline U_{20} & u_{21} & U_{22} \end{array} \right), \left(\begin{array}{c} z_T \\ z_B \end{array} \right) \rightarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array} \right)$
<u>$\zeta_1 := \zeta_1 - u_{12}^T z_2$</u>
<u>$\zeta_1 := \zeta_1 / v_{11}$</u>
$\left(\begin{array}{c c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c c c} U_{00} & u_{01} & U_{02} \\ \hline u_{10}^T & v_{11} & u_{12}^T \\ \hline U_{20} & u_{21} & U_{22} \end{array} \right), \left(\begin{array}{c} z_T \\ z_B \end{array} \right) \leftarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array} \right)$
endwhile

Figure 5.6.2.14. Upper triangular solve Variant 2

Cost of triangular solve Starting with an $n \times n$ (upper or lower) triangular matrix T , solving $Tx = b$ requires approximately n^2 flops.

Provided the solution of $Ax = b$ yields some accuracy in the solution, that accuracy can be improved through a process known as **iterative refinement**.

- Let $\hat{x}$ is an approximate solution to $Ax = b$.
- Let $\hat{\delta x}$ is an approximate solution to $A\hat{\delta x} = b - A\hat{x}$,
- Then $\hat{x} + \hat{\delta x}$, is an improved approximation.
- This process can be repeated until the accuracy in the computed solution is as good as warranted by the conditioning of A and the accuracy in b .

Definition 5.6.2.15. Hermitian positive definite matrix. A matrix $A \in \mathbb{C}^{n \times n}$ is Hermitian positive definite (HPD) if and only if it is Hermitian ($A^H = A$) and for all nonzero vectors $x \in \mathbb{C}^n$ it is the case that $x^H A x > 0$. If in addition $A \in \mathbb{R}^{n \times n}$ then A is said to be symmetric positive definite (SPD).

Some insights regarding HPD matrices:

- B has linearly independent columns if and only if $A = B^H B$ is HPD.
- A diagonal matrix has only positive values on its diagonal if and only if it is HPD.
- If A is HPD, then its diagonal elements are all real-valued and positive.
- If $A = \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right)$, where A_{TL} is square, is HPD, then A_{TL} and A_{BR} are HPD.

Theorem 5.6.2.16. Cholesky Factorization Theorem. *Given an HPD matrix A there exists a lower triangular matrix L such that $A = LL^H$. If the diagonal elements of L are restricted to be positive, L is unique.*

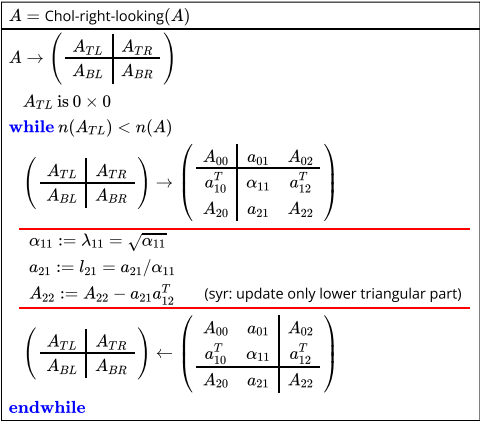


Figure 5.6.2.17. Cholesky factorization algorithm (right-looking variant). The operation ‘syr’ refers to ‘symmetric rank-1 update’, which performs a rank-1 update, updating only the lower triangular part of the matrix in this algorithm.

Lemma 5.6.2.18. *Let $A = \left(\begin{array}{c|c} \alpha_{11} & a_{21}^H \\ \hline a_{21} & A_{22} \end{array}\right), \in \mathbb{C}^{n \times n}$ be HPD and $l_{21} = a_{21} / \sqrt{\alpha_{11}}$. Then $A_{22} - l_{21} l_{21}^H$ is HPD.*

Let $\hat{x} \in \mathbb{C}^n$ equal the solution to the linear least-squares (LLS) problem

$$\|b - A\hat{x}\|_2 = \min_{x \in \mathbb{C}^n} \|b - Ax\|_2, \tag{5.6.1}$$

where A has linearly independent columns, equals the solution to the normal equations

$$\underbrace{A^H A}_B \hat{x} = \underbrace{A^H b}_y.$$

This solution can be computed via the steps

- Form $B = A^H A$. Cost: approximately mn^2 flops.
- Factor $B = LL^H$ (Cholesky factorization). Cost: approximately $n^3/3$ flops.
- Compute $y = A^H b$. Cost: approximately $2mn$ flops.
- Solve $Lz = y$. Cost: approximately n^2 flops.
- Solve $L^H \hat{x} = z$. Cost: approximately n^2 flops.

for a total of, approximately, $mn^2 + n^3/3$ flops.